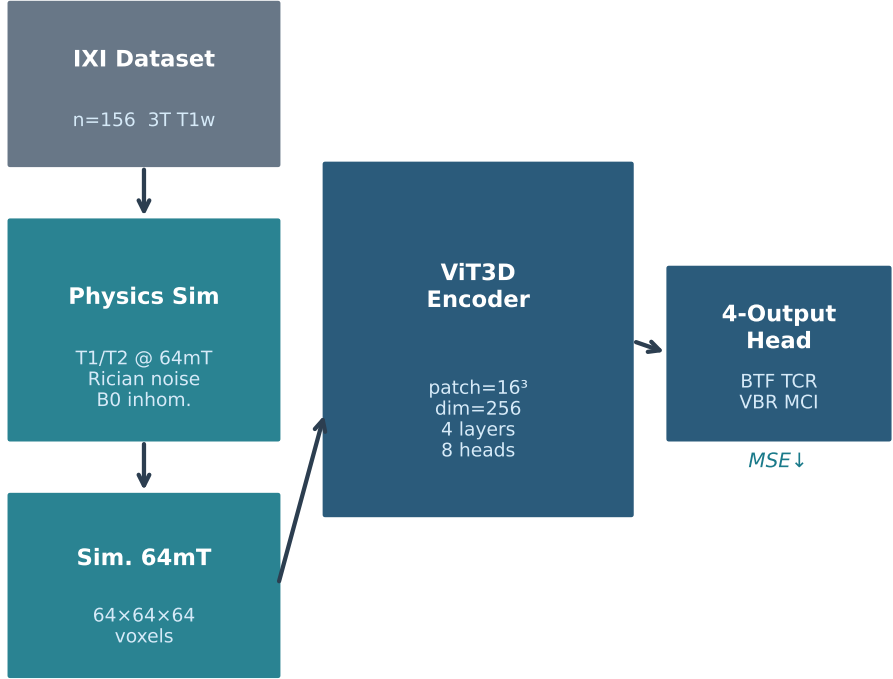


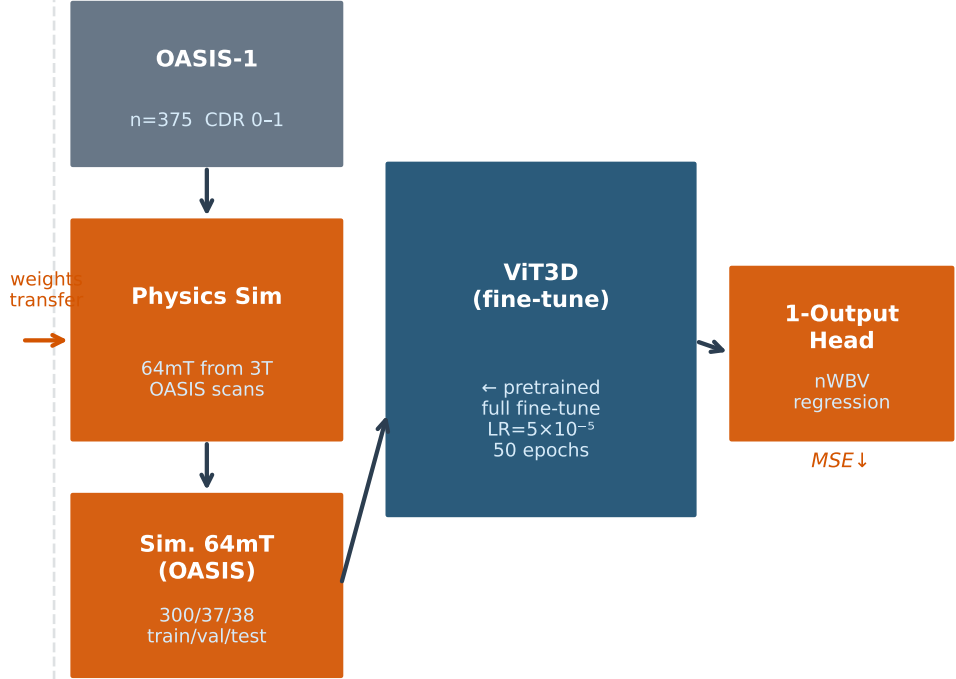
# Physics-Constrained ViT3D: Two-Stage Pipeline for nWBV Prediction from 64 mT MRI

## STAGE 1 Pre-training



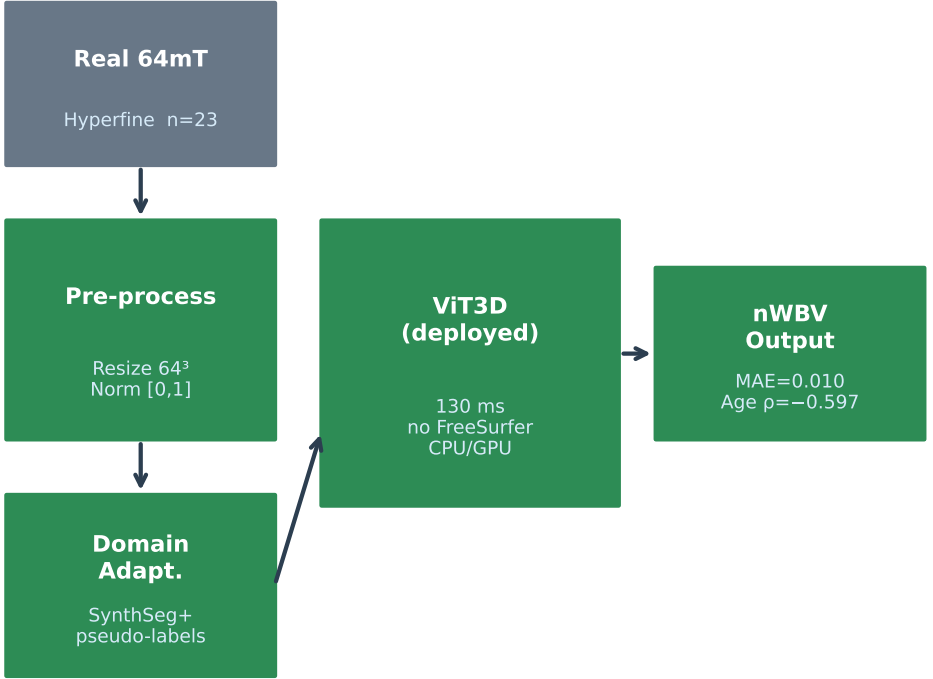
Stage 1: Physics-informed pre-training on IXI (n=156)

## STAGE 2 Fine-tuning



Stage 2: Clinical fine-tuning on OASIS-1 → r=0.892 [CI: 0.801-0.943]

## DEPLOYMENT



Deployment: 130 ms inference, no FreeSurfer dependency